

Below tasks are performed for selected countries which are Canada, Denmark, United States and Wales.

Task D.1:

For any two given months (i.e., you can assume any two months, e.g., April 2022 and May 2022), list the months, the total number of vaccines administered in each observation month in each of all countries, and the difference between the administered vaccines. Each row in the result set must have the following structure.

Note: OM2 is after OM1.

Observation Months 1 (OM1)	Country Name (CN)	Administered Vaccine on OM1 (VOM1)	Observation Months 2 (OM2)	Administered Vaccine on OM2 (VOM2)	Difference of totals (VOM1 - VOM2)
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Figure 1: Column Headers in the Result Set for Task D.1

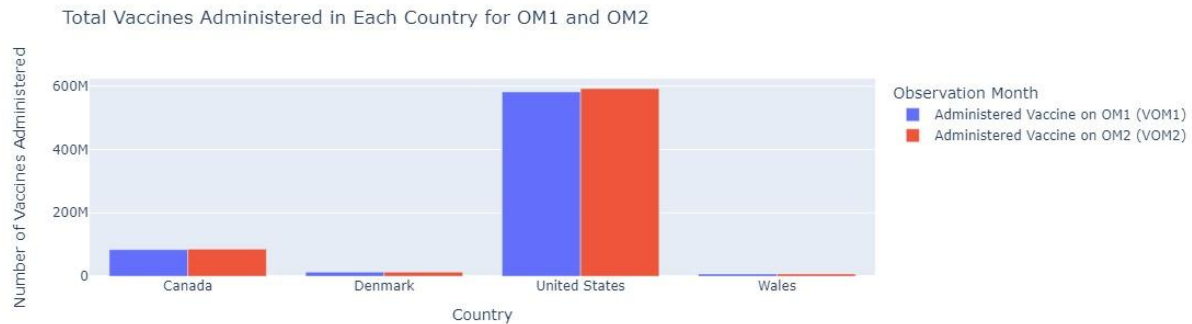
SQL Query:

```
SELECT
    '2022-04' AS "Observation Months 1 (OM1)",
    loc.location_name AS "Country Name (CN)",
    MAX(CASE
        WHEN strftime('%Y-%m', vd.date) = '2022-04'
        THEN vd.total_vaccinations ELSE 0 END)
    AS "Administered Vaccine on OM1 (VOM1)",
    '2022-05' AS "Observation Months 2 (OM2)",
    MAX(CASE
        WHEN strftime('%Y-%m', vd.date) = '2022-05'
        THEN vd.total_vaccinations ELSE 0 END)
    AS "Administered Vaccine on OM2 (VOM2)",
    MAX(CASE
        WHEN strftime('%Y-%m', vd.date) = '2022-05'
        THEN vd.total_vaccinations ELSE 0 END) -
    MAX(CASE
        WHEN strftime('%Y-%m', vd.date) = '2022-04'
        THEN vd.total_vaccinations ELSE 0 END)
    AS "Difference of totals (VOM1 - VOM2)"
FROM VaccinationData vd
JOIN Location loc ON vd.iso_code = loc.iso_code
WHERE vd.iso_code IN ('CAN', 'DNK', 'USA', 'OWID_WLS')
GROUP BY loc.location_name;
```

SQL Results:

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Data Visualization:



Task D.2:

Find the countries with more than the average cumulative numbers of COVID-19 doses administered by each country in each month.

Note: The result may include multiple countries or a single country.

Produce a result set containing the name of each country and the cumulative number of doses administered in that country. Each row in the result set must have the following structure.

Country Name	Month	Cumulative Doses
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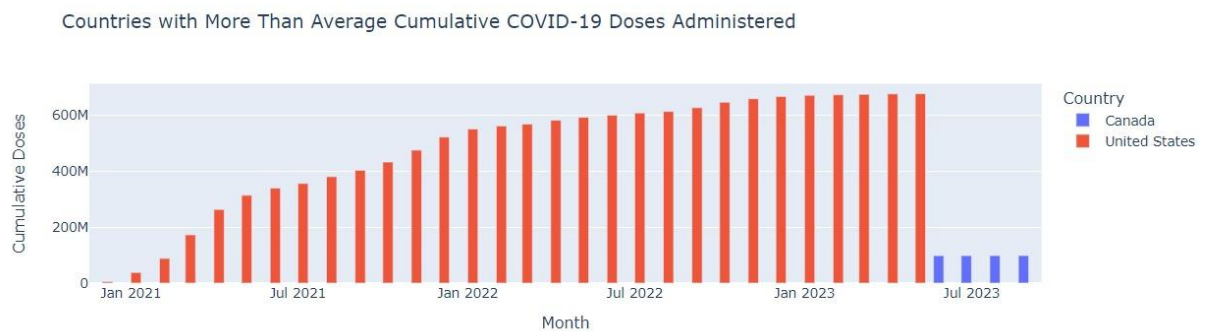
Figure 2: Column Headers in the Result Set for Task D.2

```
WITH TotalMonthlyVaccinations AS (  
    SELECT  
        loc.location_name AS country_name,  
        strftime('%Y-%m', v.date) AS month_year,  
        MAX(v.total_vaccinations) AS total_vaccinations  
    FROM VaccinationData v  
    JOIN Location loc ON v.iso_code = loc.iso_code  
    WHERE v.iso_code IN ('CAN', 'DNK', 'USA', 'OWID_WLS')  
    GROUP BY loc.location_name, month_year  
)  
Average AS (  
    SELECT  
        month_year,  
        AVG(total_vaccinations) AS average_vaccinations  
    FROM TotalMonthlyVaccinations  
    GROUP BY month_year  
)  
SELECT  
    m.country_name AS "Country Name",  
    m.month_year AS "Month",  
    m.total_vaccinations AS "Cumulative Doses"  
FROM TotalMonthlyVaccinations m  
JOIN Average avg ON m.month_year = avg.month_year  
WHERE m.total_vaccinations > avg.average_vaccinations;
```

SQL Results:

Grid view Form view			
1 Total rows loaded: 34			
	Country Name	Month	Cumulative Doses
1	Canada	2023-06	98238922
2	Canada	2023-07	98310269
3	Canada	2023-08	98365911
4	Canada	2023-09	98410188
5	United States	2020-12	5716312
6	United States	2021-01	38066296
7	United States	2021-02	88232689
8	United States	2021-03	172494725

Data Visualization:



Task D.3:

Produce a list of vaccine types with the countries taking each vaccine type. For a vaccine type that has been taken in multiple countries, the result set is required to show several tuples reporting each country in a separate tuple. Each row in the result set must have the following structure.

Vaccine Type	Country
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Figure 3: Column Headers in the Result Set for Task D.3

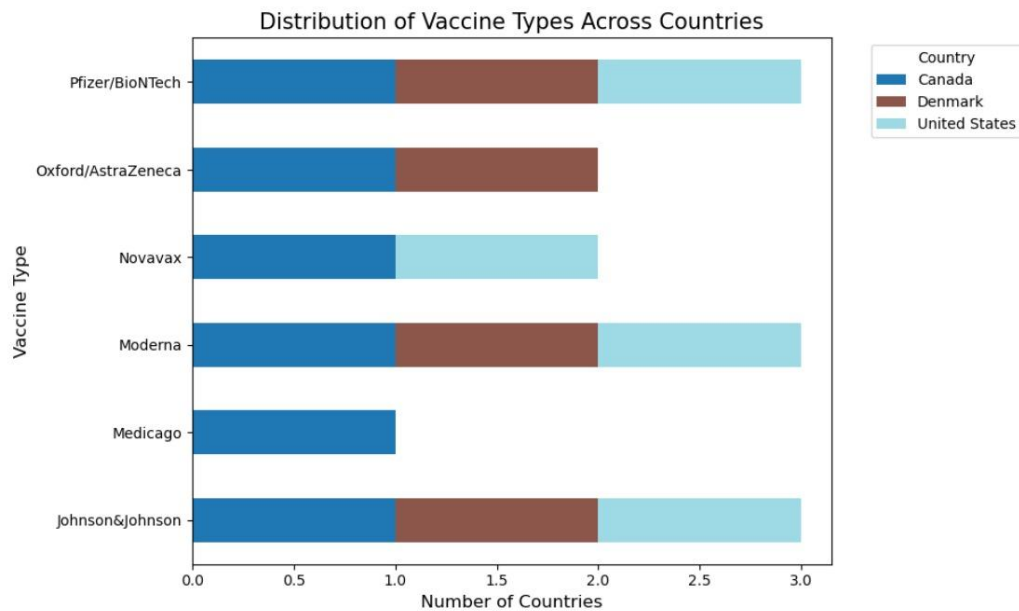
SQL Query:

```
SELECT DISTINCT
    v.vaccine_name AS "Vaccine Type",
    loc.location_name AS "Country"
FROM Vaccine v
JOIN CountryVaccinationByManufacturer cvm
    ON v.vaccine_name =
    cvm.vaccine_name
JOIN Location loc ON cvm.iso_code =
    loc.iso_code
WHERE loc.iso_code IN ('CAN', 'DNK', 'USA',
'OWID_WLS')
ORDER BY v.vaccine_name, loc.location_name;
```

SQL Results:

Grid view		Form view
		Total rows loaded: 14
Vaccine Type	Country	
1 Johnson&Johnson	Canada	
2 Johnson&Johnson	Denmark	
3 Johnson&Johnson	United States	
4 Medicago	Canada	
5 Moderna	Canada	
6 Moderna	Denmark	
7 Moderna	United States	
8 Novavax	Canada	
9 Novavax	United States	

Data Visualization:



Task D.4:

There are different data sources used to produce the dataset. Produce a report showing the total number of vaccines administered in each country according to each data source (i.e., each unique URL). Order the result set by the total number of administered vaccines. Each row in the result set must have the following structure.

Country Name	Source Name (URL)	Total Administered Vaccines
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Figure 4: Column Headers in the Result Set for Task D.4

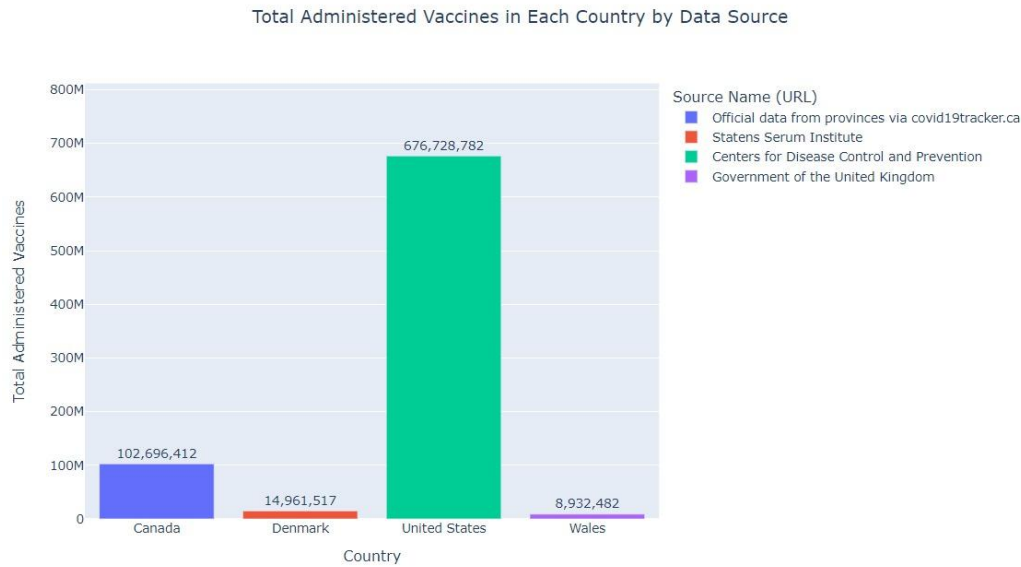
SQL Query:

```
SELECT
    loc.location_name AS "Country Name",
    s.source_name AS "Source Name (URL)",
    MAX(v.total_vaccinations) AS "Total Administered Vaccines"
FROM VaccinationData v
JOIN Location loc ON v.iso_code = loc.iso_code
JOIN DataSource ds ON v.iso_code = ds.iso_code
JOIN Source s ON ds.source_id = s.source_id
WHERE loc.iso_code IN ('CAN', 'DNK', 'USA', 'OWID_WLS')
GROUP BY loc.location_name, s.source_id
ORDER BY
    MAX(v.total_vaccinations) DESC, loc.location_name,
    s.source_name;
```

SQL Results:

Grid view Form view			
     1    Total rows loaded: 4			
	Country Name	Source Name (URL)	Total Administered Vaccines
1	United States	Centers for Disease Control and Prevention	676728782
2	Canada	Official data from provinces via covid19tracker.ca	102696412
3	Denmark	Statens Serum Institute	14961517
4	Wales	Government of the United Kingdom	8932482

Data Visualization:



Task D.5:

How do various countries compare in the speed of their vaccine administration?

Produce a report that lists all the observation months in 2022 and 2023, and then for each month, list the total number of people fully vaccinated in each one of the 4 countries used in this assignment.

Date Range (Months)	United States	Wales	Canada	Denmark
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Figure 5: Column Headers in the Result Set for Task D.5

SQL Query:

```
SELECT
    strftime('%Y-%m', vd.date) AS "Date Range(Months)",
    MAX(CASE WHEN vd.iso_code = 'USA' THEN vd.people_fully_vaccinated ELSE 0 END) AS "United States",
    MAX(CASE WHEN vd.iso_code = 'OWID_WLS' THEN vd.people_fully_vaccinated ELSE 0 END) AS "Wales",
    MAX(CASE WHEN vd.iso_code = 'CAN' THEN vd.people_fully_vaccinated ELSE 0 END) AS "Canada",
    MAX(CASE WHEN vd.iso_code = 'DNK' THEN vd.people_fully_vaccinated ELSE 0 END) AS "Denmark"
FROM VaccinationData vd
WHERE vd.date BETWEEN '2022-01-01' AND '2023-12-31'
GROUP BY strftime('%Y-%m', vd.date)
ORDER BY vd.date;
```

SQL Results:

